Md Asif Sajeed

Prospective PhD Student — Systems Security & Machine Learning

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Academic Profile

Computer Science & Engineering graduate (RUET, CGPA 3.79/4.00, ranked 14th) working at the intersection of machine learning and systems security. First author of a journal paper under review on the temporal robustness and explanation stability of graph neural network malware detectors, and co-author of three IEEE conference papers in applied deep learning. Hands-on reverse engineering, malware analysis and DFIR experience from international CTFs.

Education

Rajshahi University of Engineering & Technology (RUET) Mar 2022 – Aug 2026

B.Sc. in Engineering, Computer Science & Engineering *CGPA 3.79 / 4.00 (Rank 14)*

- Ranked **14th** in the graduating cohort. Bangladeshi 4.00 scale; 3.79 corresponds to a First Class with Distinction.
- **Thesis:** *Temporal Generalization and Explanation Stability of Graph Neural Networks for Control Flow Graph-Based Malware Detection.*
  Supervisor: Dr. Md. Nazrul Islam Mondal, Professor. External: Md. Azmain Yakin Srizon, Assistant Professor.
- Relevant coursework: Machine Learning, Artificial Intelligence, Digital Image Processing, Computer Security, Algorithms, Data Structures.

Gazipur Cantonment College 2018 – 2020

Higher Secondary Certificate (HSC), Science *GPA 5.00 / 5.00*

Publications

Journal article under review.

1. **Md. Asif Sajeed**, M. N. I. Mondal, and M. A. H. Akash, "Temporal Generalization and Explanation Stability of Control Flow Graph Neural Networks for Malware Detection," *under review, Journal of Information Security and Applications*, 2026. Preprint: arXiv:2609.24280. Code: github.com/Ho9pe/TG-CFG

Three peer-reviewed papers in IEEE conference proceedings.

1. **Md. Asif Sajeed**, S. Afroge, M. A. H. Akash, A. Khayer, and A. Al Mamun, "Memory-Efficient Deep Learning Architectures for Water Body Segmentation in Satellite Imagery: A Comparative Analysis and Ensemble Approach," *2025 28th Int. Conf. on Computer and Information Technology (ICCIT)*, Cox's Bazar, Bangladesh, Dec. 2025. doi:10.1109/ICCIT68739.2025.11491634
2. A. Khayer, S. Afroge, A. Al Mamun, **Md. Asif Sajeed**, and M. A. H. Akash, "Transformer-Based Multi-Label Disease Prediction from Radiology Reports," *2025 28th Int. Conf. on Computer and Information Technology (ICCIT)*, Cox's Bazar, Bangladesh, Dec. 2025. doi:10.1109/ICCIT68739.2025.11491786
3. M. A. H. Akash, S. Afroge, **Md. Asif Sajeed**, M. Barkullah, and A. Khayer, "An Explainable Deep Learning Framework for Multi-Class Alzheimer's Diagnosis With Multi-Scale CNN," *2026 IEEE 2nd Int. Conf. on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, Chittagong, Bangladesh, Apr. 2026. doi:10.1109/QPAIN69676.2026.11545771

Research Experience

Undergraduate Thesis — Graph Neural Networks for Malware Detection RUET, 2025 – 2026

Supervisor: Dr. Md. Nazrul Islam Mondal, Professor *Python, PyTorch Geometric, angr*

- First-author journal paper under review (JISA); preprint arXiv:2609.24280, code at github.com/Ho9pe/TG-CFG.
- Built a fully static analysis pipeline extracting control flow graphs from 1,989 Windows PE binaries (MalwareBazaar, GitHub releases) with `angr`, encoding each basic block as a 37-dimensional feature vector.
- Compared twelve GNN variants (GCN, GAT, GIN, GraphSAGE, PNA, DeeperGCN, custom CFGNet/CFGFormer models) and a flat LightGBM control under a strict temporal split — trained on 2024–2025 binaries, scored once on a disjoint 2026 corpus. Operator choice significantly affected robustness (Friedman $p = 0.0006$); PNA held 2026 AUC 0.879.
- Showed a rank reversal: the topology-free control was among the best in-distribution (AUC 0.932) but among the worst after the boundary (0.807). Neither recalibration nor ensembling substituted for operator choice; a derived architecture (CFGNet-v2) tied the best searched operator without search.

- Applied GNNExplainer across the boundary: attributions did not shift, but cross-architecture agreement sat at chance level (top-10% Jaccard 0.055 vs. 0.053 random).

Undergraduate Research Group Lead RUET, 2024 – 2026

Supervisor: Shyla Afroge

- Led a four-member research group through four CNN- and NLP-based projects, three of which reached publication in IEEE conference proceedings, coordinating experimental design, division of work, and manuscript preparation.

Professional Experience

Dynamic Solution Innovators (DSi) Dhaka, Bangladesh, Mar 2025

Industrial Attachment — Software Engineering (14 days)

- Designed the Dynamic Feedback Management System (DFMS), a role-based feedback platform with OTP-verified submission, automated approver assignment, and a multi-stage approval workflow.
- Delivered implementation-ready specifications: functional and non-functional requirements, use case and component diagrams, a normalised database schema, UI designs, and a five-sprint delivery plan.

Information System Analysis & Design — Robi Axiata PLC RUET, 2025

Systems Analyst, four-member team

- Analysed the information systems of a national telecom operator: feasibility study, stakeholder questionnaires, data flow modelling of the existing system, weighted candidate-system evaluation, and cost-benefit analysis with break-even and ROI projections.

Technical Skills

Machine Learning: Graph Neural Networks, Deep Learning, Explainable AI, Natural Language Processing, Computer Vision, PyTorch, PyTorch Geometric, scikit-learn, LightGBM, NumPy, Pandas

Security: Malware Analysis, Reverse Engineering, Static Binary Analysis, Digital Forensics, Ghidra, angr, Wireshark, Volatility 3

Programming: Python, C, C++, x86/x64 Assembly, JavaScript, PHP, SQL

Software & Tools: React, Next.js, Node.js, Laravel, MongoDB, Git, Docker, Linux, LaTeX, Figma

Awards & Competitions

- Ranked **14th** of graduating cohort, B.Sc. in Computer Science & Engineering, RUET.
- **Holmes CTF 2025** (Hack The Box, international) — solved all 65/65 challenges, placing 251st of 7,085 teams worldwide; top 4%.
- **Cyber Apocalypse CTF 2026** (Hack The Box, international) — 120/136 challenges, 210th of 6,744 teams; top 4%. Five international Hack The Box competitions completed, 2024–2026.

Selected Projects

Nibedito — Full-Stack E-Commerce Platform Next.js, Node.js, MongoDB

- Full-stack storefront with JWT authentication, bcrypt hashing, email/phone verification, and an admin dashboard; API hardened with rate limiting and server-side validation.

CollabTask — Collaborative Task Management Laravel, Tailwind CSS

- Task management application with collaborative boards.

Certifications

- **Machine Learning Specialization** — DeepLearning.AI & Stanford University, Mar 2025.
- **Introduction to Graph Neural Networks** — ApX Machine Learning, Aug 2026.
- **Data Science Math Skills** — Duke University, Feb 2025.